

Rishi More

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Summary

ML systems engineer focused on efficient and reliable LLM inference, spanning CUDA/Nsight kernel optimization, fused INT4 decoding, calibrated test-time compute, and post-training evaluation. **Co-first author of an ICML 2026 main-track paper** on distribution-free risk control for reducing reasoning compute under finite-sample error guarantees.

Experience

JHU Data Science and AI Institute (DSAI)

May 2025 – June 2026

Graduate Research Assistant, LLM Inference Optimization

Baltimore, Maryland

- Co-first authored *Conformal Thinking (ICML 2026 main track)*, introducing finite-sample calibrated early stopping for reasoning LLMs while controlling a target error rate.
- Built a vLLM-based calibration and evaluation pipeline across **4 reasoning models, 4 benchmarks, and 2.7K+ problems**, maintaining target error rates under dataset and response-length shift with finite-sample UCB calibration.
- Designed dual-threshold and ensembled halt policies that reduced reasoning tokens by **12% at matched accuracy**, with reusable tooling for calibration, evaluation, and ablation studies.

JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

Graduate Researcher, Retrieval Systems

Baltimore, Maryland

- Built a temporal RAG pipeline over **10K+ timestamped documents** using time-decay weighting and metadata-scoped indexing to prioritize current evidence over stale matches.
- Improved retrieval precision by **23%** over a dense-embedding baseline by down-weighting outdated passages in rapidly changing topics.
- Reduced stale and hallucination-prone retrievals by **31%** using reference-free consistency scoring over retrieved context.

Mastek

November 2022 – July 2023

Machine Learning Engineer Intern

Mumbai, India

- Built and deployed an on-premises edge voice-authentication system with **<100 ms p99 latency** and **96.6% accuracy**, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment.
- Engineered a TensorFlow data pipeline over **1M+ voice samples**, including spectral-gating denoising and robust feature extraction for noisy environments.
- Improved authentication reliability by **20%** in high-noise production conditions through preprocessing and feature-extraction upgrades.

Projects

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | CUDA C++, Nsight, vLLM

- Profiled batch-1 decode for **Qwen2.5-1.5B** at CUDA-kernel granularity, finding **~81%** of kernel time in weight-streaming GEMVs operating near **95%** of the measured **~250 GB/s** memory roofline.
- Built a correctness-gated fused INT4 dequantization + GEMV kernel that reached **86% of peak bandwidth**, cut memory traffic by **~8.7×** versus a two-op baseline, and improved batch-1 latency by **2.38×**.

Post-Training Failure Evaluation Harness | Python, PyTorch, DPO, LLM-as-Judge

- Built a model-agnostic harness for **6 failure modes** across a Qwen3-8B Base→SFT→DPO chain and a Qwen3-4B-Instruct comparator, evaluating **142 curated prompts**.
- Implemented position-swapped pairwise judging, log-probability calibration, and per-example audit trails; found that **10% of DPO preference wins** were factually degraded.
- Attributed regressions to specific optimization stages: DPO increased sycophancy from **34.1% to 44.9%**, while SFT raised harmful compliance from **0% to 12.5%**.

Scaffolded Multi-Agent Training for Language Agents | Python, LoRA, asyncio, BM25

- Built an asynchronous training system in which a frozen **Qwen3-235B** caregiver taught LoRA-tuned **Qwen3-8B** agents through modular memory and salience-gated BM25 retrieval.
- Across **160 training tasks** and **3 seeds**, caregiver conditions achieved **99.8–100% completion** versus **81.5–84.4%** for Solo/Peer baselines while requiring **26% fewer turns**.
- Diagnosed limited independent transfer on **40 held-out tasks (68.2–70.8% across conditions)**, isolating caregiver scaffolding rather than learned policy transfer as the source of training gains.

Education

Johns Hopkins University

May 2026

M.S.E. in Computer Science

GPA: 3.9/4.0

University of Mumbai

July 2024

B.Tech. in Computer Science, Honors in AI/ML

CGPA: 9.85/10

Technical Skills

Languages: Python, C++, Java, C, SQL

ML/LLM Systems: PyTorch, TensorFlow, Hugging Face, vLLM, LoRA, DPO, RAG, calibration, conformal/risk control

Inference & Infrastructure: CUDA C++, Nsight Compute/Systems, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark

Data & Scientific Computing: PostgreSQL, MongoDB, Pandas, NumPy, scikit-learn